

Dental caries

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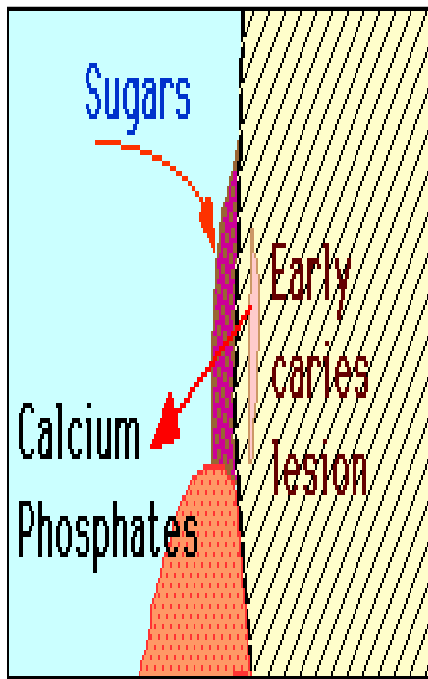
Dental caries

Definition:

- It's a progressive bacterial damage of hard tooth structure exposed to the oral environment, characterized by demineralization of the inorganic portion followed by destruction of the organic substance of the tooth leading to a cavity.

Tooth + Bacteria + CHO $\longrightarrow$ Organic acid $\longrightarrow$ Demineralization of inorganic portion followed by breakdown of organic matrix.

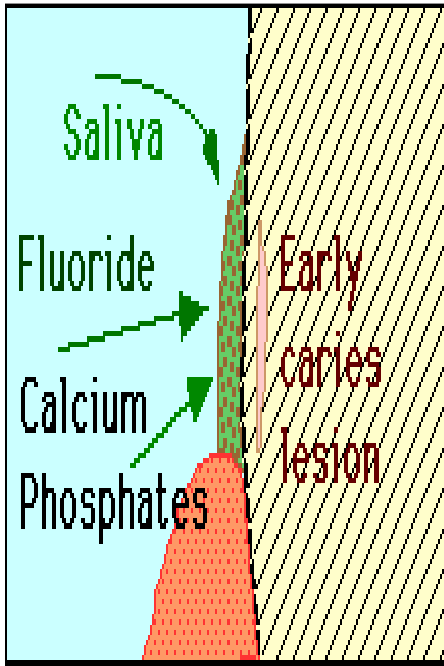
- Dental caries is considered to be a dynamic process consisting of alternating periods of destruction (**demineralization**) and repair (**remineralization**), occurring over a period of time.



Demineralization:

When carbohydrates undergo fermentation by the bacteria, they **form acids** which start to **dissolve** the enamel

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- an early caries lesion occurs due to loss of Calcium and Phosphates

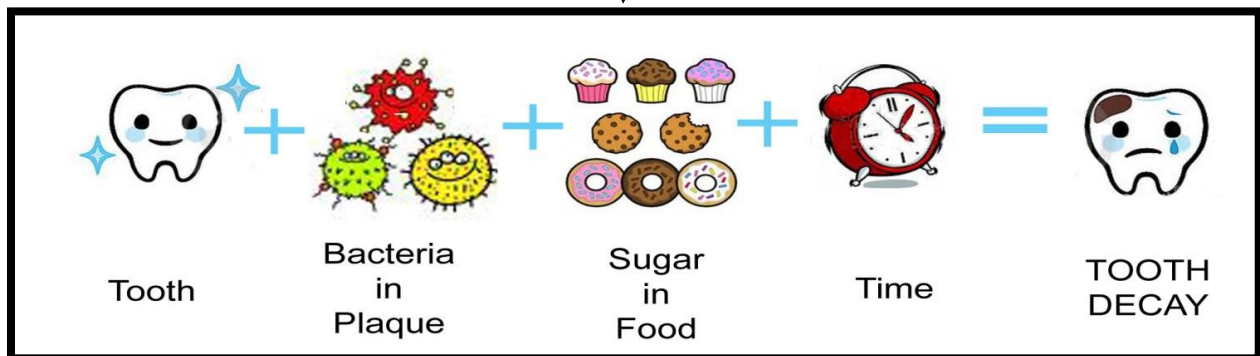


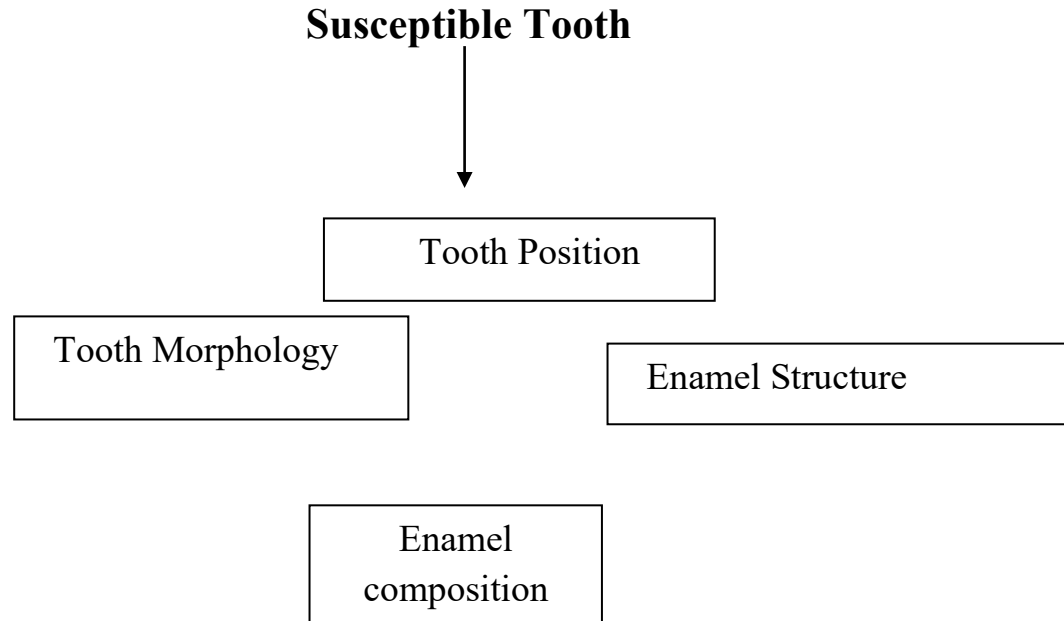
Remineralization:

When sugar consumption **has ceased**, saliva can wash away sugars and buffer the acids.

Calcium and Phosphates can again enter the tooth.
The process is strongly facilitated by fluorides

Etiology of dental caries





1- Tooth Position:

- Upper teeth are more susceptible to caries than lower teeth due to salivary action.
- Posterior teeth are more susceptible than anterior teeth due to the presence of pits and fissures.
- Malposed and crowded teeth and teeth related to clasps and orthodontic appliances are more susceptible to caries because of the creation of stagnation areas.

2- Tooth Morphology:

(a) Pit and fissure caries:

- Increased number of fissures increases caries.
- Increased depth of fissures increases caries.
- Shape of fissure: I- and inverted Y-shaped fissures are more susceptible to caries than U- and V-shaped fissures.

(b) Smooth surface caries:

- Interproximal surfaces: The contact areas are more susceptible to caries.

- Buccal and lingual surfaces: The cervical third of buccal and lingual surfaces are more susceptible to caries.

3- Enamel Structures:

- Enamel hypoplasia and enamel hypocalcification do not affect the initiation of caries but affect the progression of caries compared to normal teeth.

4- Tooth Composition (Role of fluoride in dental caries):

1- Decreasing demineralization process: Fluoride replaces the hydroxyl group in the hydroxyapatite forming fluorapatite which is less soluble in acids.

2- Increasing remineralization process.

3- Inhibition of bacterial growth.

Role of Saliva in dental caries:

(1) Saliva has a washing effect on the teeth.

(2) Saliva contains bicarbonates and phosphates acting as a buffering effect.

(3) The acquired enamel pellicle which is derived from salivary glycoprotein protects enamel surfaces from being colonized by oral bacteria.

(4) Saliva contains immunoglobulin A(IgA) which plays a role in,

a. Killing or inhibition of bacterial growth.

b. Prevention of adherence of bacteria to tooth surface.

(5) Saliva contains antibacterial substances: These include lysozymes, peroxidase and lactoferrin.

Microorganisms:

- ✓ It is generally agreed that dental caries is caused by acid resulting from action of microorganisms on carbohydrates.
- ✓ The most important organism in dental caries is **Streptococcus mutans.**

- ✓ **S. Mutans** has a role in initiation of dental caries, while **Lactobacillus** is associated with the progression of dental caries.

The characteristic features of Cariogenic bacteria:

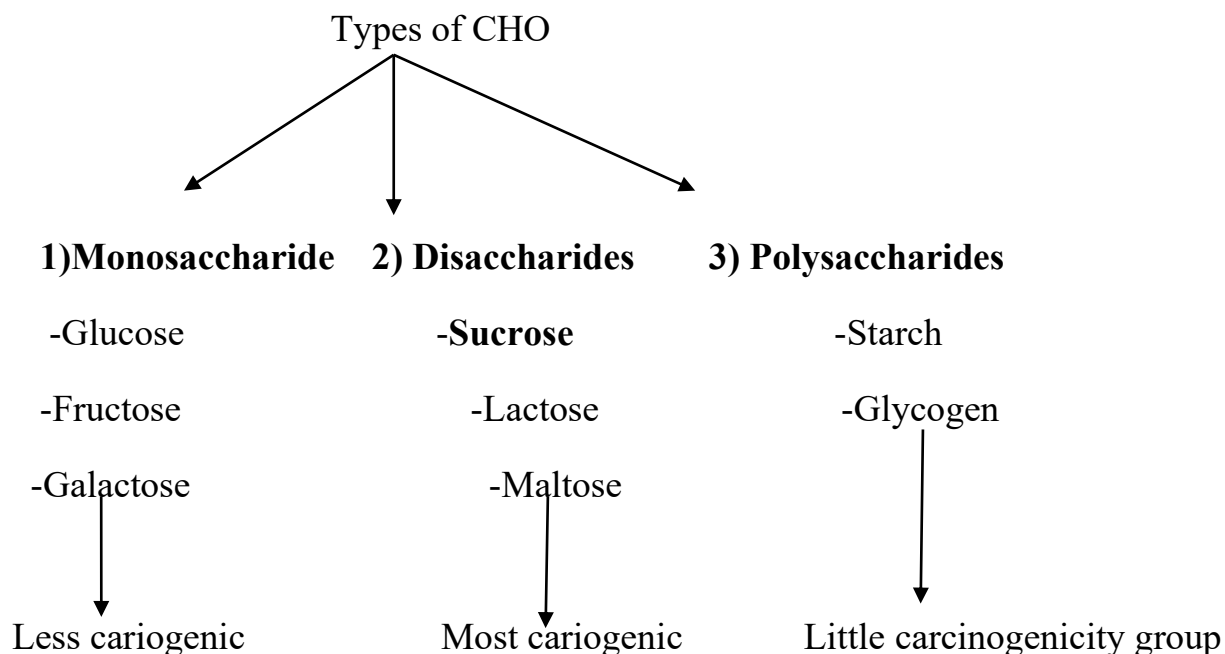
1-**Acidogenic property:** This means that the bacteria have the ability to produce organic acid rapidly from fermentable carbohydrates.

2- **Aciduric property:** This means that the bacteria can live and grow in high acidic media, i.e., low pH levels.

3- **Formation of abundant extracellular polysaccharides,** especially dextran and Levan so these bacteria can produce caries.

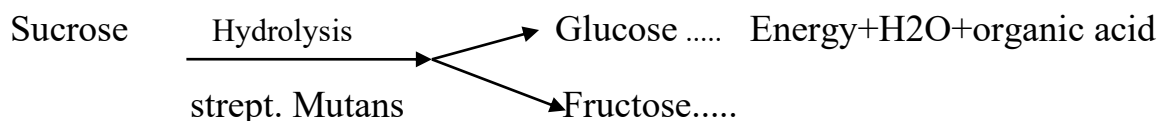
Carbohydrates (CHO):

- ✓ CHO are the only component of the diet which can be acted upon by microorganisms in the oral cavity. Proteins and fats undergo little or no change in the mouth.

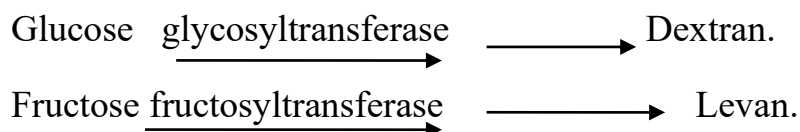


Sucrose is considered the arch-criminal of dental caries due to

- Sucrose is of low molecular weight and can diffuse rapidly into the dental plaque.
- Sucrose is of low cost, so it is the most common and most frequently consumed carbohydrate (table sugar, drinks, jam, etc.).
- Sucrose is broken down rapidly by the bacteria with the production of organic acids.



- Sucrose, with the help of cariogenic bacteria (Strept. mutans) plays an active role in the building of dental plaque by forming extracellular polysaccharides, mainly Dextran, which forms the main bulk of dental plaque and Levan, which acts as a store for CHO).



Dextran

- Extracellular polysaccharide.
- Glucose polymer.
- Adhesive.
- Forms the main bulk of dental plaque.

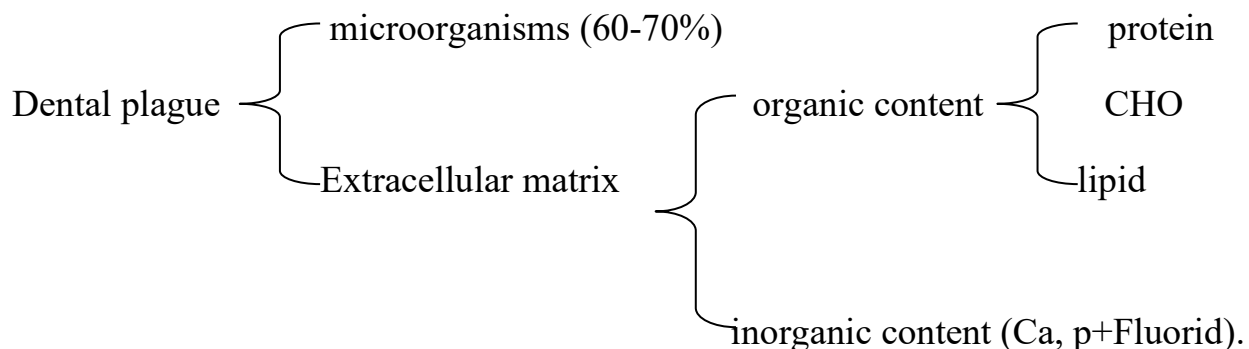
Levan

- Extracellular polysaccharide.
- Fructose polymer.
- Less adhesive.
- Acts as storage.

Dental plaque

Definition: A biofilm that builds up on and around the teeth. Plaque is Sticky and usually colorless. It forms every day on teeth.

Composition:



Notes:

- Proteins derived from saliva and gingival fluids include (glycoprotein, amylase, desquamated epithelial cells and Ag).
- CHO which include extracellular polysaccharide(Dextran+Levan).

Mechanism of formation of dental plaque:

1- Acquired enamel pellicle:

- This form is a glycoprotein that is derived from the saliva and adsorbed on the cleaned tooth surface.

Function of enamel pellicle:

- Protect the tooth from being colonized by oral bacteria.
- It is on this component of dental plaque takes place; thus, the pellicle serves as a nutrient for plaque microorganisms.



2- Colonization and growth phase:

Colonization phase

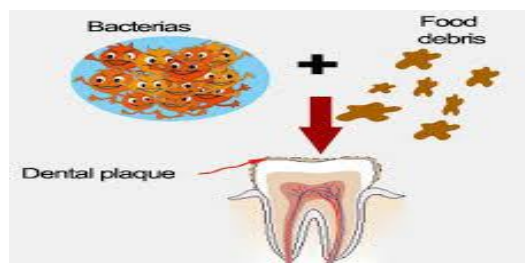
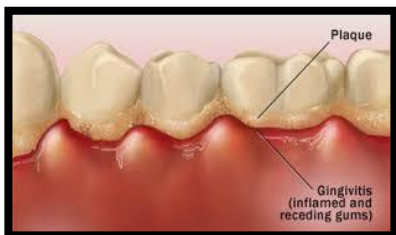
- Bacterial adhesion and colonization on the tooth surface begin within minutes after the pellicle is formed.
- The predominant initial colonizers (pioneer bacteria) are *S.sanguis*. *S. oralis*.
- It's believed that some of these pioneer organisms release an extracellular enzyme capable of liberating carbohydrate from pellicle glycoprotein and using it as so a source of energy.

Growth phase (8-48 hours):

- This begins with the ingress and proliferation of secondary invaders *Streptococcus mutans*.
- ***S. mutans*** appears in large numbers and has two effects,
 - 1- Fermentation of sucrose and production of organic acid.
 - 2- Synthesis of extracellular polysaccharides (dextrane & Levan) which form the main bulk of dental plaque.
- 3- **Maturation (2 weeks):** Maturation of dental plaque by further colonization of other bacteria. Also, there is buildup of plaque substance by polysaccharides which are produced by microorganisms.

Role of Dental Plaque

- 1- Acting as a diffusion-limiting membrane by
 - Delayed diffusion of salivary buffers into the plaque and
 - Delayed diffusion of acids outward into the saliva.
- 2- Contributes to adhesiveness, bulk and resistance to washing.



Histopathology Features of Enamel Caries

When enamel loss minerals and caries progress, the enamel develops several zones visible under a light microscope.

1- Translucent zone:

- This zone lies at the advancing front of the enamel lesion.
- This zone is slightly more porous than sound enamel, having a pore volume of 1% compared with 0.1% of sound enamel.

2- Dark zone:

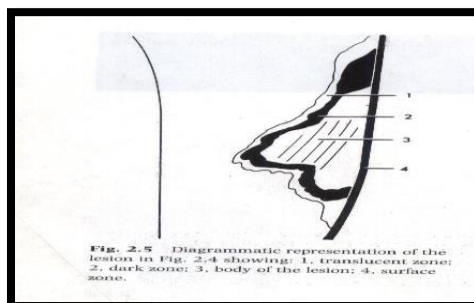
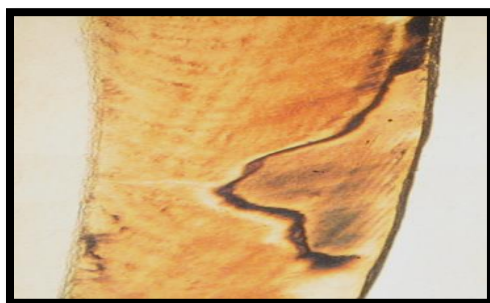
- Lies superficial to the translucent zone.
- Called positive zone as it is always present.
- Pore volume is 2-4%; increased porosity in this zone is due to the degree of demineralization.

3- Body of lesion:

- Forms the bulk of the lesion and lies between the dark zone and the surface zone.
- Area of greater demineralization, having a pore volume of 5-25%.

4-Surface zone:

- Relatively intact.
- Zone of greater mineralization (high fluoride) and pore volume of only 1%.



Histopathology of Dentine Caries

- ✓ Once bacteria penetrate enamel, they spread laterally along the DEJ and attack dentine over a wide area. Dentine becomes infected through dentinal tubules, which provide an easy pathway for bacteria.

Five zones that have been recognized in dentine caries include:

1- **Zone of Dentinal sclerosis:** characterized by deposition of calcium salts in dentinal tubules, which finally seals them against further penetration by microorganisms.

2- Zone of decalcification:

- Lies in front of bacterial invasion.
- Only a few tubules are invaded by bacteria (**pioneer bacteria**) even before clinical evidence of caries.

3- Zone of bacterial invasion:

- Dentinal tubules are packed with the invading bacteria, leading to their confluence.
- Coalescence and breakdown of adjacent tubules lead to formation of "liquefaction foci".

4- **Zone of destruction:** The liquefaction foci enlarge and increase in number, resulting in cracks or **clefts** containing bacteria and necrotic dentine.



Fig. 2.12 Pioneer organisms at the advancing front of the zone of bacterial invasion in dentine caries.



Fig. 2.13 Packing of the dentinal tubules by bacteria in the zone of bacterial invasion (Gram stain).

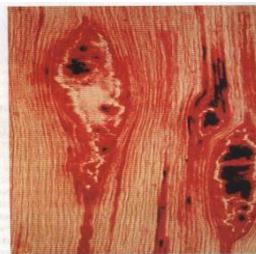


Fig. 2.14 Liquefaction foci in dentine caries.



5-Zone of reactionary or tertiary dentin: